

NICK RODRIGUEZ

Founding Engineer | Full-Stack TypeScript & Python | AI Integration

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Problem solver who builds software end-to-end. For **5+ years** has shipped production systems independently, including **News Nexus** (a 5-service platform delivering weekly news-clip reports under a US federal contract) and **Kyber Vision** (volleyball-analytics mobile app for a European sports-tech product). Prior 15 years as a federal economist (US Government, ICF, HLB Decision Economics, EIA). Open to founding-engineer and early-stage roles where breadth, ownership, and shipping matter more than specialization.

TECHNICAL SKILLS

Frontend React, React Native (Expo), Next.js, Redux Toolkit, Tailwind

Backend Node.js (ExpressJS), Python (Flask, FastAPI), REST API design, JWT auth

Data Postgres + Sequelize at production scale, schema migration (SQLite → Postgres)

Queues & Infra BullMQ + Redis, custom FIFO with cooperative cancellation

AI Integration OpenAI, HuggingFace, sentence-transformers (semantic + dedupe scoring)

Media Pipelines ffmpeg, YouTube API, Playwright scraping

DevOps Ubuntu, reverse proxy, GitHub

Solo Ownership architecture → build → ops → stakeholders

EXPERIENCE

Software Engineer • **Freelance**

Nov 2020 – Present

Building production software end-to-end, independently, for paying clients, equity-stake partners, and personal products. Stack spans full-stack web, mobile, AI integration, and DevOps.

- ▶ **News Nexus.** (paid client work) Sole builder and operator of a 5-service production platform supporting a US federal contract with the Consumer Product Safety Commission via Kinetic Metrics. Ingests news from multiple sources into Postgres 16 via a shared Sequelize model package; runs deduplication, semantic and location scoring, OpenAI state classification, and content scraping through Node and Python workers; serves a Next.js review portal for human-in-the-loop clip approval and weekly PDF/Excel contract delivery. Completed SQLite → Postgres migration.
- ▶ **Tastermonial.** (pro bono trial engagement) Partnered with the founder to add Oura sleep tracking across their stack: Elixir (Phoenix) and Python (FastAPI) backends, new PostgreSQL schemas, Junction webhook ingestion, and Flutter mobile UI. Stood up an isolated parallel ecosystem (DB, both APIs, mobile) outside Tastermonial's prod/dev environments and shipped an auth-enabled Flutter web build so test users could trial it safely.
- ▶ **Kyber Vision.** (equity engagement) Built the React Native mobile app for a European volleyball analytics product (kybervision.eu). Handles auth, team and player admin, live match scripting via a custom court-zone tap and swipe-wheel gesture system, video upload and review, action-to-video timeline sync, and highlight montage requests. Integrates with an Express API, shared Sequelize models, and a BullMQ + Redis + ffmpeg worker that uploads to YouTube and generates montages.
- ▶ **What Sticks.** (what-sticks.com) Sleep and weather tracking web app. Daily API calls pull Oura Ring sleep scores and weather conditions; computes per-user temperature-to-sleep correlations over time.

Economist • **US Federal Government**

2012 – 2020

Eight years as a federal economist focused on supply-chain analytics, price-growth analysis, and vendor proposal evaluation. Team lead on multiple engagements. Tools: regression, uncertainty analysis, forecasting, BLS price-index methodology, VBA, SAS JMP.

- ▶ **Ship Construction Price-Growth Negotiation.** (lead analyst) Built the Government's analytical position for a major proposal negotiation. For each part, ran regressions against historical purchases, mapped to a BLS price index via qualitative-similarity and growth-range checks, combined third-party BLS forecasts into a Government-position forecast, and produced uncertainty bounds. Implemented in VBA and SAS JMP.
- ▶ **NATO Inflation History.** Authored a 40-year history of US inflation covering drivers and policy context; presented at a NATO conference.
- ▶ **VBA Tooling for External Data Ingest.** Built Excel/VBA tools that automated extraction from external sources to feed recurring team analyses, replacing manual workflows.

Associate · ICF International

Jan 2009 – Jan 2012

Three years in ICF's energy practice delivering forecasting and analytics for US electricity market clients including utilities, RTOs, and federal agencies. Tools: linear programming, econometric/regression modeling, large database management.

- ▶ **US Electricity Market Forecasting.** Used ICF's proprietary LP model of US electricity markets to deliver power-plant valuations, energy-price forecasts, and market-share forecasts for utility and RTO clients.
- ▶ **NREL Cross-Model Comparison Study.** Lead on database development and major contributor to the published report comparing technology costs and performance across seven energy market models (MARKAL, NEMS, GPRA, ReEDS, MERGE, MiniCAM, EPA IPM).
- ▶ **ARRA Federal Grants Advisory.** Advised regional governments on applying for ARRA funding for energy-efficiency infrastructure including public transit and building upgrades.

Economist · HDR | HLB Decision Economics

Nov 2007 – Dec 2008

Economic and risk modeling for infrastructure clients across the US, spanning public-sector evaluation and private-sector market assessment. Tools: cost-benefit analysis, Monte Carlo cost-risk methods, regression for market sizing.

- ▶ **Southeastern US Construction Market Assessment.** For a large US contractor, assessed market potential across water/wastewater, power, seaports, rail, and airports. Led areas involving risk and statistical modeling. Output shaped the client's resource-allocation strategy.
- ▶ **Transportation Infrastructure Project Evaluations.** Cost-benefit modeling for public-entity clients; reports informed how clients prioritized public funding awards.
- ▶ **Cost and Schedule Risk Modeling.** Built Monte Carlo cost-risk analyses estimating total cost and approximate schedule under uncertainty, used to size contingency and set baselines.

Industry Economist · US Department of Energy / EIA

Jun 2005 – Nov 2007

Industry economist at the US Energy Information Administration focused on retail electricity. Tools: survey-data methodology and edit-check design, energy-model interpretation, web data-product development, analytical writing.

- ▶ **Electricity Sales and Revenue Survey.** Developed and analyzed edit checks, improving data-collection efficiency and surfacing trends for EIA reports.
- ▶ **OAIF Detail.** Summarized energy-model results during a detail focused on economic effects of a potential new policy.
- ▶ **EIA Web Data Products.** Developed, improved, and maintained EIA's retail electricity web data products, working with IT to take products to production. Led projects.

EDUCATION

La Capsule · Certified Full-Stack Web & Mobile Developer (JavaScript)

Sep 2024 – Dec 2024

George Mason University · M.A., Economics

2005 – 2007

George Mason University · B.A., Economics

1999 – 2004

INTERESTS

Guitar · Reading · French